VetLink Animal Decision Support System

Comprehensive Technical Audit, Readiness Evaluation & Path-to-Defense Action Plan

Institution: College of Computing Studies, EARIST | **Date:** October 23, 2026
Project: A Deep Learning Animal Decision Support System Using MobileNetV3 and Grad-CAM for VetLink Clinic
Defense Window: November 9–11, 2026 | **Current Codebase Readiness:** **82%** (Target: 100% by Oct 25)
Overall Capstone Progress: **60%** (Code + Beneficiary Testing + Chapters 1–5 Draft)

1. Executive Summary & Objective Status

This audit establishes an unsparing distinction between **software system readiness (82%)** and **overall capstone completion (60%)**. While the core AI classification engine (11 classes across 4 species) and clinical safety logic are fully operational, three specific visual and structural flaws must be resolved before presenting the system to the faculty adviser on October 29.

Module / Component	Readiness	Status	Audit Findings & Technical Gaps
1. AI Model & WASM Engine	**92%**	SOLID	11 classes, 4 species, 3.1 MB INT8 TFLite, 38–45ms client-side WASM inference; zero cloud cost.
2. CDSS Clinical Intelligence	**92%**	SOLID	Comorbidity detection (delta ≤ 55%), discrete Laplacian blur rejection, Grad-CAM, in-chart override, printable Rx.
3. Database & Offline Sync	**82%**	GOOD	IndexedDB multi-store schema, conflict resolution UI, offline auth; needs canvas downsampling to stop image bloat.
4. Frontend UI & Native PWA	**70%**	ACTION REQ.	"Install PWA" button has no onClick event handler; home dashboard displays static hardcoded numbers.
TOTAL SOFTWARE SYSTEM	**82%**	**SOLID B**	**Core engine is proven; 3 frontend code fixes needed to achieve 100% production readiness.**

2. Technical Inventory: What is Currently Working in the System

A. Edge Machine Learning & Explainable AI

- **Edge Model (vetlink_model_v2.tflite):** MobileNetV3-Large fine-tuned across 11 classes spanning Dogs, Cats, Birds, and Rabbits. Quantized to 3.1 MB INT8, executing client-side in 38–45 ms via browser WebAssembly (WASM).
- **Mathematical Blur Rejection:** Discrete 2D Laplacian convolution kernel ($\nabla^2 f$) with variance threshold ($\sigma^2 < 80.0$) blocking out-of-focus captures before inference.
- **Erythema-Weighted Grad-CAM:** Physiological Erythema Index (2R – G – B) with border suppression highlighting true lesion boundaries.

B. Clinical Decision Support System (CDSS)

- **Calibrated 60% Decision Boundary:** Confidence ≥ 60% auto-populates care protocols; < 60% triggers mandatory laboratory scraping/cytology prompts.
- **Suspected Comorbidity Detection:** Detects co-occurring conditions (e.g. Hotspots + Ringworm) when prediction delta ≤ 55% or secondary ≥ 20%.
- **In-Chart Diagnostic Override:** Dropdown allowing attending vets to reassign diagnoses, ensuring human clinical governance.
- **Printable Rx Slip:** Print layout (@media print) generating official veterinary prescription slips with doctor signature lines.
- **Longitudinal Recheck:** Dual-column Baseline vs. Recheck comparison modal with 4-stage healing progress scoring.

C. Hardware Camera & Persistence

- **Live Camera Viewfinder:** Direct device video stream with macro lesion framing overlay and instant camera/gallery toggle.
- **Offline IndexedDB & Sync:** Full local schema with offline demo auth fallback and two-way conflict resolution panel.

3. The 3 Critical Deficiencies (What Needs Immediate Action)

Deficiency / Flaw	Current Code Defect	Consequence During Adviser / Panel Demo	Required Fix
1. Dead Install Button	App.jsx:212 has no onClick handler.	Adviser clicks "Install PWA" and literally nothing happens. System will be criticized as a website rather than a PWA.	Attach beforeinstallprompt event listener to trigger native installation dialog.
2. Hardcoded Dashboard	ClinicalAnalysis.jsx has static numbers: 24 Cases, 7 Pending.	When users intake 5 new patients, the home dashboard still says 24. Panels will flag the dashboard as non-functional mock data.	Query real counts dynamically from IndexedDB using getPatients() and getPendingSyncQueue().
3. Image Bloat in DB	Raw 12MP camera photos (4–8 MB) saved as full base64 strings.	IndexedDB accumulates 100+ MB after 15 consultations, causing mobile browser lag or sync timeouts over clinic Wi-Fi.	Downscale photos via HTML5 canvas to 1024x1024 JPEG (quality 0.82) before storing in IndexedDB.

4. Master Milestone Roadmap to Oral Defense (Oct 24 – Nov 11)

Date Window	Phase & Focus	Key Deliverables & Action Items	Progress
Oct 24–25 (Weekend)	**Weekend Sprint** Code Freeze	• Fix the 3 frontend items (PWA install, dynamic metrics, canvas downsampling). • Rehearse end-to-end camera intake on mobile device. • **System achieves 100% production readiness.**	**82% → 100%**
Oct 26–28 (Mon–Wed)	**Clinic Trials** VetLink Animal Clinic	• Field testing with licensed veterinarians at VetLink Animal Clinic. • Collect ISO/IEC 25010 & UAT survey evaluations (Mean & SD). • Gather qualitative clinician feedback for Chapter 4.	**Objective 3 Complete**
Oct 29 (Thursday)	**Adviser Demo** Sign-off	• Live demonstration of working PWA, WASM edge inference, and Rx slips. • Present preliminary clinic testing evaluation data. • **Secure signed Defense Approval Letter.**	**Approval Secured**
Oct 30 – Nov 3	**Manuscript Draft** Chapters 1–5	• Insert Table 3.1 (19,094 images) and Table 3.2 into Chapter 3. • Transcribe Classification Report and Confusion Matrix into Chapter 4. • Submit complete Chapters 1–5 draft on November 3.	**Draft Submitted**
Nov 4–8	**Defense Prep** Slide Deck	• Prepare 15-slide presentation deck focusing on multi-species, WASM, and CDSS. • Conduct team mock defense rehearsals and Q&A; practice.	**Rehearsed**
Nov 9–11	**Final Oral Defense**	• Deliver professional live clinical demo and defend capstone research.	**GRADUATION**